

Learn Home Automation with Arduino

Lesson 4: Controlling Relays & Lights

Controlling Relays & Lights – Study Notes

Key Concepts

1. **Electromechanical relay operation** – how a coil energises a switch contact.
2. **Isolation & safety** – using a relay (or opto isolator) to keep low voltage control circuitry separate from high voltage loads.
3. **Flyback (free wheel) diode** – protecting the driving transistor/Microcontroller from the inductive kick of the relay coil.
4. **Current limiting for LEDs** – choosing the right resistor to drive an LED from 5 V.
5. **Proper wiring for 120 V AC loads** – polarity independent connections, use of a rated relay contact, and basic safety practices.

Summary

A 5 V relay can be used as a bridge between a low voltage control circuit (e.g., an Arduino, Raspberry Pi, or a simple switch) and a high voltage load such as a 120 V AC lamp. The relay's coil is driven by the 5 V side; when the coil is energized, a set of contacts close, allowing mains voltage to flow to the lamp. Because the coil is an inductive load, a flyback diode must be placed across it to safely dissipate the voltage spike when the coil is turned off.

The same relay can also switch a low voltage LED. The LED is powered directly from the 5 V supply through a current limiting resistor, while the relay's contacts act as a simple on/off switch. By wiring the LED and the lamp to separate contacts (or using a single pole double throw (SPDT) relay), you can control both loads independently or together, depending on the circuit design. Safety is paramount when dealing with mains voltage: always use a relay rated for at least the voltage and current of the lamp, keep wiring tidy, and never work on a live circuit.

Important Points

- **Relay rating:** Choose a relay whose contact rating exceeds the lamp's voltage (120 VAC) and current (10 A for typical household lamps).
- **Drive circuit:**
 - Use an NPN transistor (or MOSFET) to switch the 5 V coil from a microcontroller pin.
 - Base resistor (10 k or a typical 2 mA base current) or calculate for your MOSFET gate.
- **Flyback diode:** Connect a diode (e.g., 1N4007) cathode to +5 V, anode to the coil's low side. It clamps the inductive spike to ~0.7 V.
- **LED resistor calculation:**

\[

$$R = \frac{V_{\text{supply}} - V_{\text{f}}}{I_{\text{LED}}}$$

\]

Example: 5 V supply, LED forward voltage 2 V, desired 20 mA !' $(R = \frac{5-2}{0.02}=150 \text{ }\Omega$

- **Wiring the lamp:**

- Connect one side of the lamp to the **line** (hot) of the mains.
- Connect the other side to the **normally open (NO)** contact of the relay.
- The **common (COM)** contact goes to the **neutral** (or the other way around—relay contacts are polarity agnostic).

- **Safety tips:**

- Disconnect mains power before wiring or modifying the circuit.
- Use a **project box** or insulated enclosure for the relay and wiring.
- Verify with a multimeter that there is no continuity between the low voltage side and the mains side when the relay is de-energised.
- Consider adding a **fuse** (e.g., 2 A) on the lamp side for extra protection.

Examples

Example 1 – Driving a 5 V LED with a Relay

Goal: Turn an LED on/off using a microcontroller pin.

Components:

- 5 V relay (coil 70 mA)
- NPN transistor (2N2222)
- 1 k Ω & 6R $\frac{1}{4}$ W 6—7F
- 1N4007 diode (flyback)
- LED + 150 Ω $\frac{1}{4}$ W 6—7F

Circuit:

1. Microcontroller pin !' 1 k Ω " & 6R $\frac{1}{4}$ W 6—7F
2. Emitter of 2N2222 !' ground.
3. Collector !' low side of relay coil.
4. Other coil terminal !' +5 V.
5. Flyback diode across coil (cathode to +5 V).
6. LED + resistor connected between +5 V and the relay's **COM** contact; the **NO** contact goes to ground.

Explanation:

When the MCU drives the base high, the transistor conducts, energising the coil. The relay contacts close, completing the LED circuit and lighting the LED. Releasing the base turns the LED off, and the diode safely dissipates the coil's inductive kick.

Example 2 – Switching a 120 V Lamp with the Same Relay

Goal: Use the same 5 V control signal to turn a mains lamp on/off.

Additional components:

- Relay rated "e /250 VAC, "e /10 A (e.g., Songle SRD 05VDC SL1).
- Inline fuse (2 A) on the lamp side.
- Small piece of heat shrink or electrical tape for insulation.

****Wiring (continuing from Example /1):****

1. ****COM**** of the relay !' one terminal of the lamp.
2. ****NO**** of the relay !' ****neutral**** (or the other hot line, depending on wiring convention).
3. The other lamp terminal !' ****line**** (hot) from the mains supply.

****Explanation:****

When the coil is energized, the COM NO contacts close, allowing mains voltage to flow through the lamp, lighting it. When the coil is de-energised, the contacts open, cutting power to the lamp. The low voltage control circuit remains completely isolated from the 120 /V side, ensuring safety.

Quick Review

1. Why is a flyback diode required across a relay coil?
2. How do you calculate the resistor value for a 5 /V LED that has a forward voltage of 2.2 /V and a desired current of 15 /mA?
3. What relay contact rating should you select for a 120 /V, 0.5 /A lamp?
4. Describe the role of the transistor (or MOSFET) in the relay driving circuit.
5. List two safety precautions you must take when wiring the 120 /V side of the circuit.

Further Reading

- ****Opto isolators**** for completely separating control and mains circuits.
- ****Solid state relays (SSRs)**** – advantages and limitations compared to electromechanical relays.
- ****Triac based lamp dimming**** and how it differs from simple on/off switching.
- ****Electrical safety standards**** (e.g., NFPA 70 – National Electrical Code).
- ****Power MOSFET driver circuits**** for faster relay coil switching.

